Currency crises and the euro

How can balance sheets and expectations turn a funding shock into a crisis?

Today's route

Currency mismatches can reverse the trade benefits of depreciation.

Pegs can fail through deteriorating fundamentals or coordination.

A currency union changes the adjustment tools and financial risks.

LECTURE 6

Balance-sheet exposure

Why can a weaker currency reduce domestic demand?

A firm borrows dollars but earns home currency

Balance sheet, home currency	Before	After depreciation
Operating assets	200	200
USD debt: one dollar	150	180
Equity	50	20

Home per USD rises 20%, from 150 to 180. Asset values are held fixed.

Trade and financial channels can oppose each other



A currency move changes liabilities as well as product prices.

Debt and equity share bad states differently

Foreign finance	What happens in a recession?
Fixed nominal debt	Promised payment initially stays fixed
Equity ownership	Dividends and market value can absorb losses

Equity shares risk, while financing terms, control and valuation still matter.

LECTURE 6

Pegs under pressure

When does defending a fixed rate become impossible or too costly?

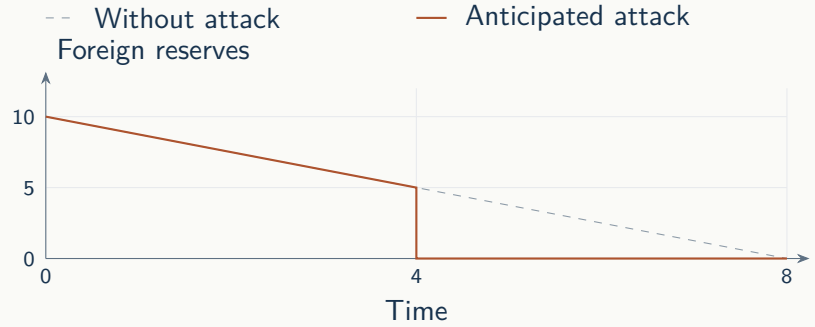
A first-generation crisis mechanism

Domestic credit keeps expanding while the peg fixes money demand.

Foreign reserves offset the expansion and decline.

The policy combination eventually cannot continue.

Reserves can be exhausted by an anticipated attack



Schematic path, not a fitted crisis date or empirical reserve series.

A second-generation crisis mechanism

Defending the peg raises interest rates and domestic economic costs.

Investors' exit expectations can increase the cost of defence.

The authorities may choose to abandon the peg.

A coordination example

Two large investors can each withdraw 4 units.

The authority can absorb 6 units of withdrawals.

One exit is manageable. Two simultaneous exits break the peg.

The coordination game

Investor 1 / Investor 2	Stay	Exit
Stay	(0, 0)	(0, -1)
Exit	(-1, 0)	(+2, +2)

Both staying and both exiting can be equilibria.

A backstop can change incentives

Credible support can remove the payoff from joining a run.

Support needs resources, legal authority and believable conditions.

A liquidity backstop does not automatically repair insolvency.

LECTURE 6

Currency unions

When is sharing one monetary policy worth the loss of local adjustment?

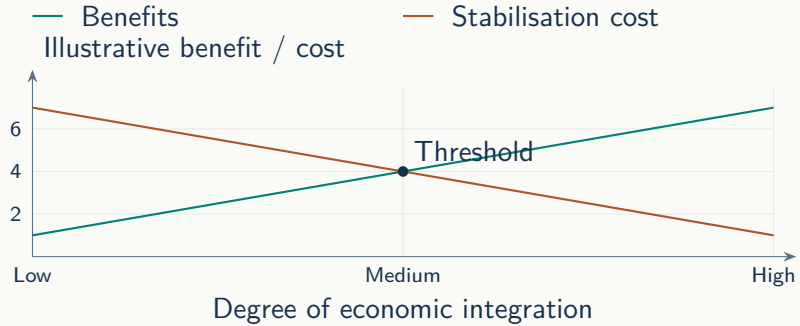
The benefits and costs of a common currency

Trade and financial integration reduce conversion costs and currency risk.

A common rate cannot respond separately to every national shock.

Other adjustment channels become more important.

An integration trade-off



A schematic OCA trade-off, not a measured threshold for any nation.

A common currency does not make every price identical

Euro-area countries share monetary policy.

Their consumption baskets, energy exposure, local taxes and price adjustments still differ.

One currency does not imply one national inflation rate.

Different baskets can produce different inflation

Hypothetical basket	Country A	Country B
Energy weight	10%	30%
Other goods/services weight	90%	70%
Energy price increase	20%	20%
Other price increase	2%	2%
Basket inflation	3.8%	7.4%

Same component price increases, different fixed basket weights.

A shared nominal rate can mean different real rates

Hypothetical comparison	Country A	Country B
Same nominal rate	4%	4%
Expected basket inflation	2%	4%
Approximate expected real rate	2%	0%

The purchasing-power cost depends on expected inflation.

Euro government bonds can also have different yields

Same currency, different issuers and credit risk.

Liquidity, collateral value and market conditions also differ.

National inflation is not a mechanical add-on to a euro sovereign yield.
The plotted ECB AAA curve is a benchmark sample, not every nation's curve.

Relative prices can adjust without a currency change

$$\Delta \ln q = \pi^{\text{foreign}} - \pi^{\text{home}} \quad \text{when } E \text{ is fixed.}$$

Higher local inflation can create real appreciation inside a currency union.

LECTURE 6

Banks and sovereigns

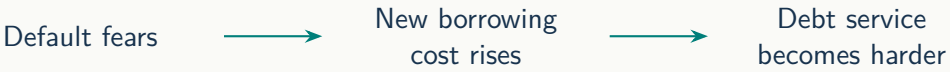
How can two balance sheets make each other weaker?

Banks transform maturities

Assets	Liabilities
Longer-term loans and securities	Deposits withdrawable on demand

A solvent bank can lack cash for simultaneous withdrawals.

A sovereign refinancing feedback



The higher funding cost can make the original fear more plausible.

The bank–sovereign loop



A historical backstop: OMT in 2012

The ECB set out its OMT technical framework on 6 September 2012.

Eligible sovereign support required programme conditionality.

The mechanism targeted severe transmission and redenomination fears.

Exercise: liquidity support or a fiscal repair?

A bank has sound long-term loans but sudden deposit withdrawals.

A government faces a temporary market closure with credible future surpluses.

Another government cannot plausibly fund its promises in real terms.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.